



CANDIDATE  
NAME

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INDEX  
NUMBER

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**BIOLOGY**  
**Paper 2: Core Paper**

**8875/02**  
**26<sup>th</sup> August 2014**  
**2 hours**

Additional Materials: Writing Paper

**READ THESE INSTRUCTIONS FIRST**

Write your index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are two sections in this paper.

**Section A]**

Answer all questions

**Section B]**

Answer **ONE** question. Answer each part on a **separate** piece of paper.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [ ] at the end of each question or part of the question.

| For Examiner's Use |    |
|--------------------|----|
| Section A          | 40 |
| 1 [10]             |    |
| 2 [12]             |    |
| 3 [12]             |    |
| 4 [6]              |    |
| Section B          | 20 |
| 5a / 6a [7]        |    |
| 5b / 6b [7]        |    |
| 5c / 6c [6]        |    |
| TOTAL              | 60 |

## Section A

Answer **all** questions in this section.

- 1 All cells are surrounded by a cell surface membrane. There are also membrane systems within the cell, as well as several types of membrane-bound organelles. Some of these have a single membrane surrounding them, while others have two.

All cell membranes have a similar basic structure but the proportions of the various molecules present vary between membranes and even between the inner and outer layers of a membrane.

Fig. 1.1 shows a generalised cell surface membrane.

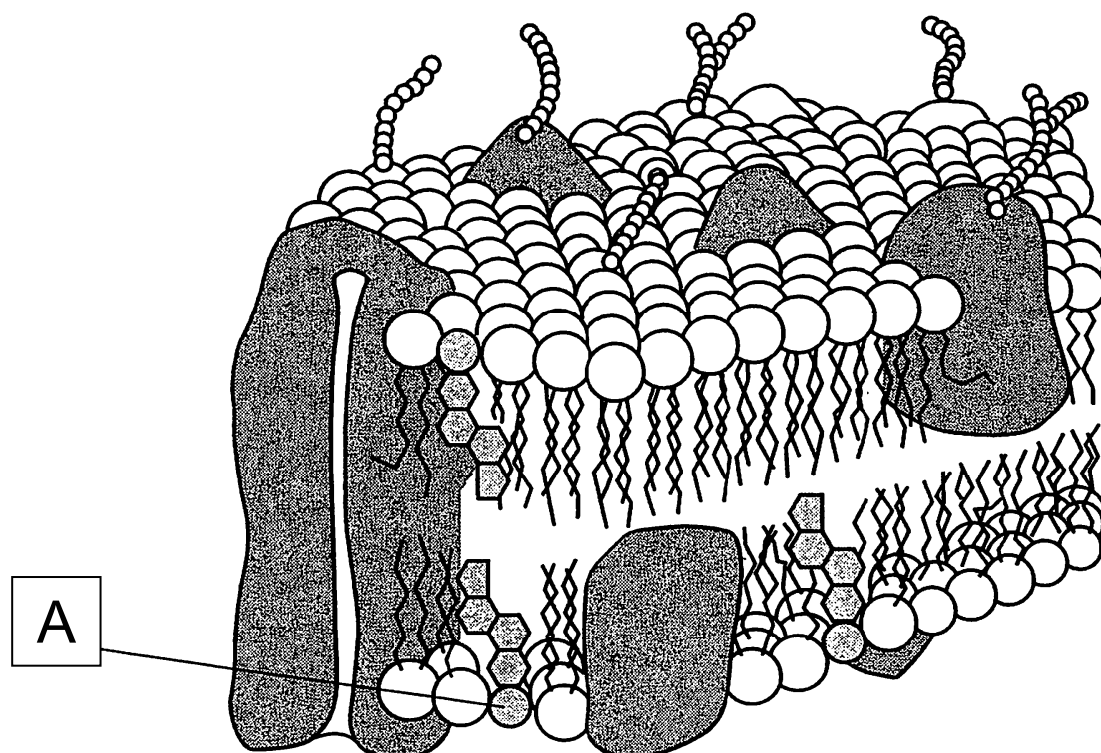


Fig. 1.1

With reference to Fig. 1.1,

- (a) (i) identify the molecule A and describe its function.

[2]

|            |                                                                                                                                  |
|------------|----------------------------------------------------------------------------------------------------------------------------------|
| Molecule A | <u>cholesterol</u>                                                                                                               |
| Function   | <u>regulates/maintains membrane fluidity.</u><br><br><u>Or</u><br><u>Ensures membrane stability at high and low temperature.</u> |

[L1]

- (ii) With reference to Fig. 1.1, state one way in which the outer surface membrane is different from the inner surface membrane. [1]

| SC                                             | OR                                       | E                                                                                                                                             |
|------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| fig 1.1<br>outer<br><br>different<br><br>inner | Presence of Glycoproteins and glycolipid | 1. presence of <u>glycoprotein</u> and <u>glycolipid</u> in the outer surface membrane where <u>none occur on the inner</u> surface membrane. |

[L2]

- (iii) state one functional property resulting from this difference [1]

| SC           | OR                                                                                                                                                                                                                     | E                                                                                                                                                                                                                                 |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| one property | <ul style="list-style-type: none"> <li>outer membrane able to merge with other cells with complementary glycoprotein and / or glycolipid matrixes</li> <li>cell to cell recognition</li> <li>cell signaling</li> </ul> | 1. <u>outer surface membrane</u> able to adhere / attach with other cells with <u>complementary glycoprotein and / or glycolipid matrixes</u><br>2. the potential for <u>cell-to-cell recognition</u><br>3. <u>cell signaling</u> |

[L2]

- (b) Table 1.1 shows the approximate percentage of protein in different membranes.

**Table 1.1**

| Organelle     | Membrane       | Protein as a percentage of dry weight. |
|---------------|----------------|----------------------------------------|
| chloroplast   | outer membrane | 54                                     |
| chloroplast   | inner membrane | 70                                     |
| mitochondrion | outer membrane | 55                                     |
| mitochondrion | inner membrane | 80                                     |

- (b)(i) using data from Table 1.1, describe one similarity in the distribution of proteins within the two membranes of chloroplast and mitochondrion and suggest a reason. [3]

|            |  |
|------------|--|
| Similarity |  |
| Reason     |  |

| SC           | OR                                                                                                                                                                                                  | E                                                                                                                                                                                                                                                                                                                                                |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Both Similar | <ul style="list-style-type: none"> <li>inner membrane had a higher occurrence of protein</li> <li>DRUM</li> <li>both had almost equal levels of protein for outer membrane</li> <li>DRUM</li> </ul> | 1. <u>Inner membranes</u> of both mitochondria and chloroplast are <u>higher in protein percentage</u> compared to their respective outer membranes. <u>Mitochondria (inner 70 vs outer 54) / Chloroplast (inner 80 vs outer 55)</u><br><br>2. <u>Outer membranes protein percentage</u> in both mitochondria and chloroplast <u>were almost</u> |

|  |  |                                                                                                               |
|--|--|---------------------------------------------------------------------------------------------------------------|
|  |  | <u>equal (mitochondria 55 vs Chloroplast 54)</u><br><br>Reason x 1 mark<br>Data cited x 1 mark<br><br>(Max 1) |
|--|--|---------------------------------------------------------------------------------------------------------------|

| SC     | OR                                                                                                                                                                                                                                                                               | E                                                                                                                                                                                                                                                                                                                                         |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reason | 1] proteins involved with ECM <ul style="list-style-type: none"> <li>enzymatic roles</li> <li>outer mainly with transport</li> </ul> 2] both had almost equal levels of protein for outer membrane <ul style="list-style-type: none"> <li>outer mainly with transport</li> </ul> | 1. Inner membrane protein percentage is <u>higher due to involvement of proteins with intra cellular matrix and enzymatic roles</u> , compared to mainly transport proteins in the outer membrane.<br>2. the <u>outer membrane protein percentage was almost identical</u> due to <u>similar roles of mainly transport</u><br><br>(Max 1) |

[L3]

(b) (ii) Describe and explain one physical way in which the inner membrane of a mitochondrion is adapted to its function.

[3]

| SC                              | OR                                                                                                                                                                                          | E                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| inner membrane adapted function | many infolds of the inner mem<br>presence of electron transport chain and ATP synthase for ATP production.<br>increase surface area- allow greater maximizing for process of ATP production | 1. <u>many folds (cristae)</u> of the inner mitochondrial membrane.<br>2. <u>presence of electron transport chain and ATP synthase for ATP production</u> .<br>3. <u>increase surface area</u> - allows more <b>elements of ETC and Stalked particles</b> to be grouped together in close proximity to one another. |

[L2]

[Total: 10]

2 Fig. 2.1 shows the elongation phase of the cellular process of transcription.

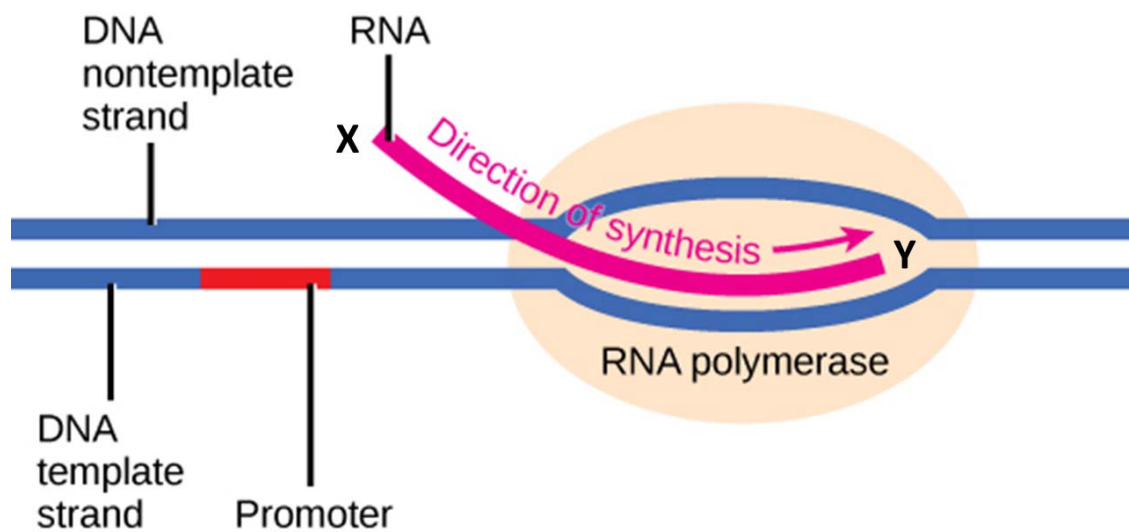


Fig. 2.1

(a)(i) Identify the directionality of the RNA transcript at ends X & Y.

[1]

X: 5'

Y: 3'

[L1]

(ii) State what are the substrates which are not shown in Fig. 2.1 that RNA polymerase allows in into its active site for the process of elongation to take place.

[1]

### Ribonucleoside triphosphates

(Reject: nucleotides/RNA nucleotides/ribonucleotides)

[L1]

(iii) The elongation of RNA transcripts requires energy. Explain how this energy is derived.

[1]

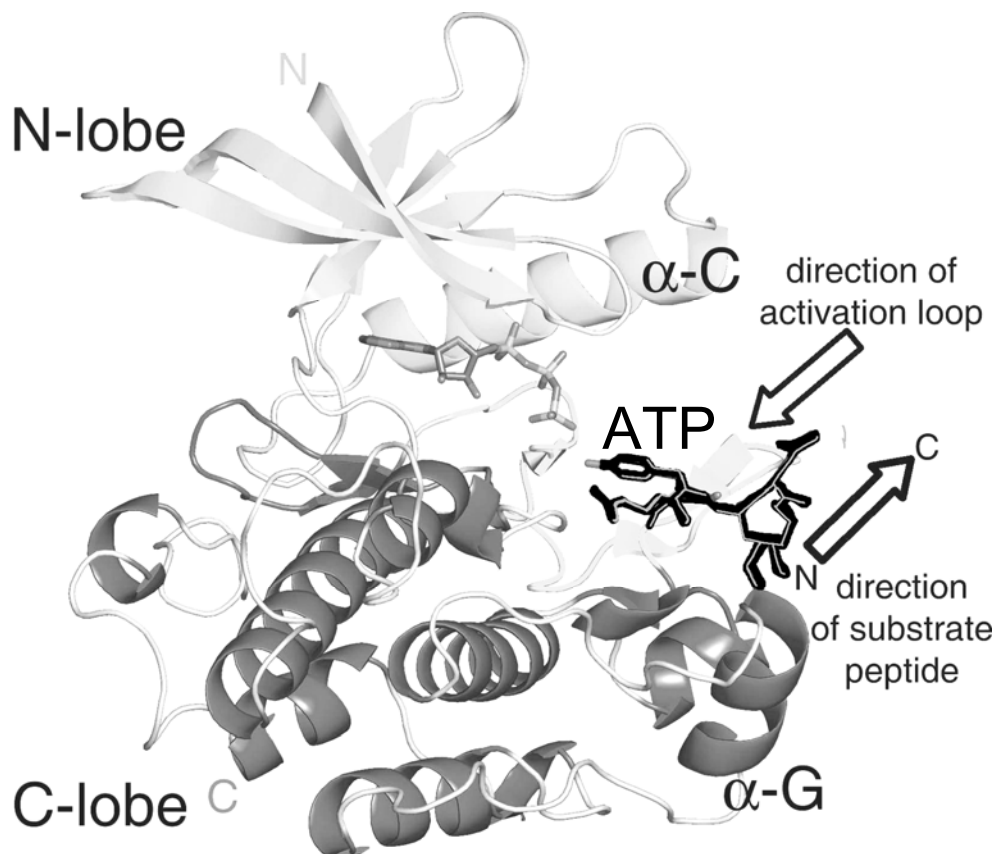
| SC                                                              | OR                                                                                                      | E                                                                                                                                                                 |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CW:<br>Explain how<br><br>QKW:<br>Energy derived, drive process | <ul style="list-style-type: none"> <li>• Energy from breaking of bonds in incoming monomers.</li> </ul> | 1. The energy to drive this process is obtained from the <u>hydrolysis of the incoming ribonucleoside triphosphate</u> molecules, which releases a pyrophosphate. |

[L1]

Protein kinases are a class of enzymes that are involved in the triggering of the cell cycle. Kinases add phosphate to molecules, and the modification can serve as a "switch" to turn events in the cell on or off.

Cyclin dependent kinases (Cdk) regulate the cell cycle.

Fig. 2.2 shows a representation of Cdk.



**Fig. 2.2** Anatomy of a Cdk catalytic domain. The substrate peptide, docked onto the activation loop, and an ATP analog are shown in black.  $\alpha$ -Helix-C, which contains residues critical for catalysis, and  $\alpha$ -helix-G, which is involved in docking protein substrates, are labeled.

(b) Describe how specialised enzymes are structurally specific to their substrate [2]

| SC                                               | OR                                                                                            | E                                                                                                                                                                                  |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enzymes<br>Structurally<br>specific<br>substrate | Globular shape<br>able to maintain a specific<br>configuration<br>complementary to substrate. | 1. enzymes are <b>globular in shape</b> and are in <b>specific configuration</b> that is <b>complementary</b> to the substrate conformation<br>2. <b>Lock and Key hypothesis</b> . |

[L1]

With reference to Fig. 2.2,

(c) explain the specific significance of the following parts of protein kinase [2]

| Feature           | Significance                                                          |
|-------------------|-----------------------------------------------------------------------|
| N-Lobe and C-Lobe | 1. maintains <b>specific globular configuration</b> of protein kinase |

|                                                     |                                                                                                                                                                                                                                  |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\alpha$ -C and $\alpha$ -G                         | Catalytic R Groups and binding of the R groups                                                                                                                                                                                   |
| directions of activation loop and substrate peptide | 2. once <u>catalytic R groups attach to substrate</u> causes a <u>strain in substrate</u> due to <u>opposing directions of the activation loop and substrate peptide</u> . this <u>lowers activation energy</u> of the reaction. |

[L3]

Cyclin-dependent kinases (Cdk) are a family of protein kinases first discovered for their role in regulating the cell cycle.

(d) Suggest how Cdk may be regulated in its action

[1]

| SC                       | OR                                                                                        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Protein kinase regulated | Enzyme competitive inhibition<br>restriction of substrate<br>substrate as limiting factor | <ol style="list-style-type: none"> <li>1. Cdk are enzymes therefore regulation of protein kinase can be via <u>competitive inhibition</u></li> <li>2. Regulation of protein kinase via <u>restriction of substrate</u>.</li> <li>3. <b>Cdk active only when complexed with cyclins</b></li> </ol> <p><a href="http://java-srv1.mpi-cbg.de/publications/getDocument.html?id=ff8080812e8f3dae012e900ec661011a">http://java-srv1.mpi-cbg.de/publications/getDocument.html?id=ff8080812e8f3dae012e900ec661011a</a></p> |

(Max 1)

[L3]

Cdk works always in conjunction with cyclins (another cofactor in the regulation of the Cell cycle) the complexes that are formed when these two are combined makes Cdk active.

Fig. 2.3 shows the relative levels of active Cdk (protein kinase activity) during the cell cycle.

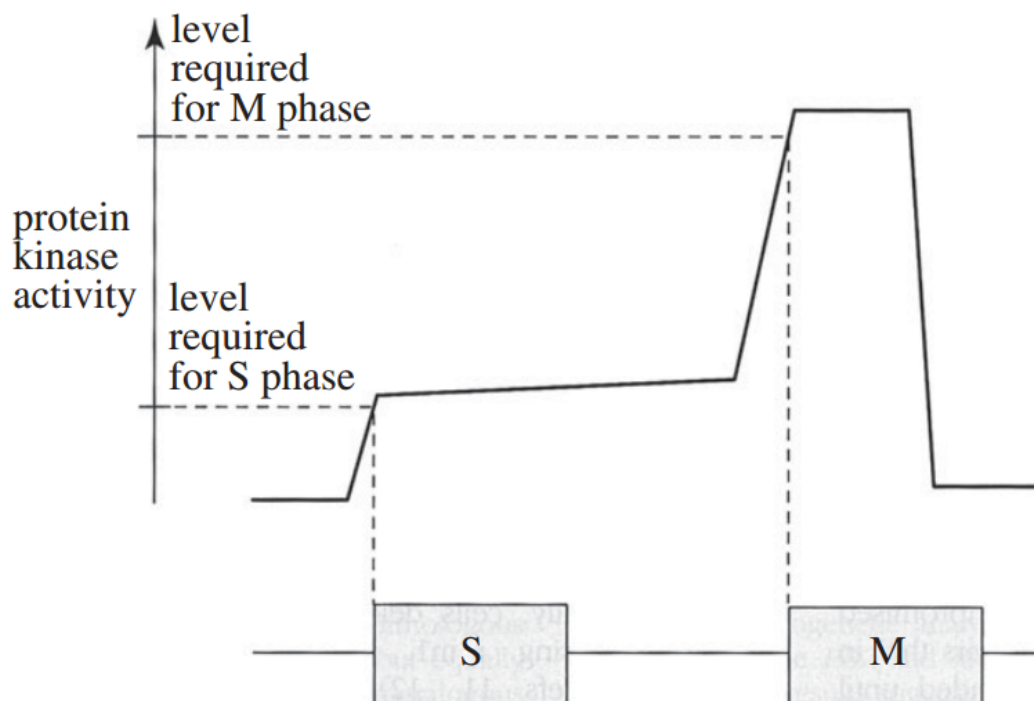


Fig 2.3

(e) Explain what takes place in the gap in-between S phase and M (cell division)

[2]

| SC                        | OR                                                                                                                                                                  | E                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cell Cycle<br>Lag between | <b>G2 phase</b> <ul style="list-style-type: none"> <li>• rapid cell growth</li> <li>• checking of Replication</li> <li>• repair</li> <li>• G2 checkpoint</li> </ul> | 1. The gap represents the <b>G2 phase</b> in cell cycle where <b>rapid cell growth takes place</b><br>2. in G2 there is the <b>checking on the process of replication and repair</b><br>3. in G2 <b>if replication is successful</b> then it <b>passes the G2 checkpoint</b> which is the control point in cell division.<br><br><div style="text-align: right;">(Max 2)</div> |
| S phase                   | <b>interphase</b> <ul style="list-style-type: none"> <li>• DNA replication</li> </ul>                                                                               |                                                                                                                                                                                                                                                                                                                                                                                |
| M Phase                   | <b>Cell division</b> <ul style="list-style-type: none"> <li>• PMAT</li> </ul>                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                |

[L2]

(f) Explain how CDK is used in the regulation of the cell cycle

[2]

| SC                          | OR                                                                                                                                                                                                                                                               | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1]CDK                       | <b>1] CDK</b> <ul style="list-style-type: none"> <li>• precursor / activator for each stage</li> <li>• is needed to <b>preserve the stage</b></li> </ul>                                                                                                         | 1. CDKs control the <b>initiation of the cell cycle</b> , they are <b>precursors / activators</b> to each stage of the cell cycle and must be present in critical amounts as indicated by the <b>threshold limits</b> for <b>S phase</b> and a <b>higher threshold limit</b> for <b>Mitotic / meiotic / cell division</b> (OWTTE)<br>2. Presence of <b>Cdk above respective threshold levels</b> are required for both <b>S and M phases</b> of the cell cycle<br>3. <b>to allow time for the intermediate processes between S phase and M phase to complete.</b><br><br><div style="text-align: right;">(Max 2)</div> |
| 2] Regulation<br>Cell cycle | <b>2] Control</b> <ul style="list-style-type: none"> <li>• <b>Threshold levels</b></li> <li>• indicate control over cell division, takes place only if <b>threshold met</b> DRUM</li> <li>• <b>maintained during phase-</b> DRUM-second level control</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

[L3]

[Total: 12]

3 Fig. 3.1 shows the plasmid pEOD.

Plasmid pEOD is a cloning vector used in creating recombinant DNA. It contains two selectable marker genes, Kan<sup>R</sup> (kanamycin resistance gene) and Amp<sup>R</sup> (ampicillin resistance gene). A multiple cloning site (MCS) is located within the Kan<sup>R</sup> gene.

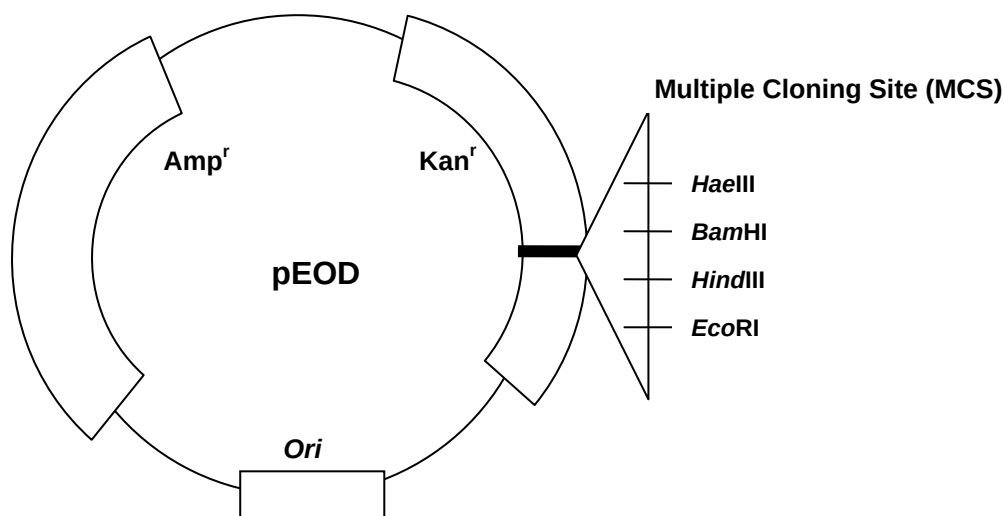


Fig. 3.1

A target sequence of human DNA is replicated using the Polymerase Chain Reaction (PCR).

The target DNA sequence is shown in Fig. 3.2.

5' - GGATCCATCCCGATCGATCCTTTAAGCTAGCTAGGATCC - 3'  
3' - CCTAGGTAGGGCTAGCTAGGAAATTCGATCGATCCTAGG - 5'

Fig. 3.2

(a) Design two 8-base long primers for the sequence to be replicated.

[2]

| SC                         | OR                                                                                                                                       | E                                                                                                                                                           |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Target sequence of primers | <ul style="list-style-type: none"> <li>target 8 bases at 3'end</li> <li>anti-parallel</li> <li>2 primers: forward and reverse</li> </ul> | Primer 1: <u>5' – GGATCCAT – 3'</u><br>Primer 2: <u>3' – ATCCTAGG – 5'</u>                                                                                  |
| Sequence of primer         | <ul style="list-style-type: none"> <li>Complementary base sequences to target</li> </ul>                                                 | <ul style="list-style-type: none"> <li>5' and 3' indicated correctly on both primers - 1 mark</li> <li>correct sequence of both primers – 1 mark</li> </ul> |

[L2]

(b) State three differences between the process of PCR and natural DNA replication in cells.

[3]

| SC                                | OR (basis of comparison) | E (Any 3 of the following)                                                                                                          |
|-----------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Difference PCR Vs DNA replication | 1. Nature of primers     | DNA replication involves <u>RNA primers</u> while PCR requires <u>DNA primers</u> .                                                 |
|                                   | 2. Location              | DNA replication takes place in the <u>nucleus of the cell</u> while PCR is <u>automated</u> /takes place in a <u>thermocycler</u> . |

|  |                              |                                                                                                                                                                                   |
|--|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | 3. Enzymes involved          | DNA replication involves <u>DNA polymerase III</u> and <u>DNA polymerase I</u> while PCR involves the enzyme <u>Taq polymerase</u> .                                              |
|  | 4. Proof-reading             | In DNA replication, the <u>daughter strands</u> are <u>proofread</u> by DNA polymerase I but there is <u>no proofreading of daughter strands</u> in PCR by <u>Taq</u> polymerase. |
|  | 5. Unzipping of template DNA | In PCR, <u>high temperatures</u> are required for <u>denaturing the strands</u> while in DNA replications, the enzyme <u>helicase unwinds the strands</u> .                       |
|  | 6. Synthesis of primers      | In DNA replication, <u>primase synthesises the RNA primers</u> , but in PCR, the <u>primers are added in</u> /primase is not involved.                                            |

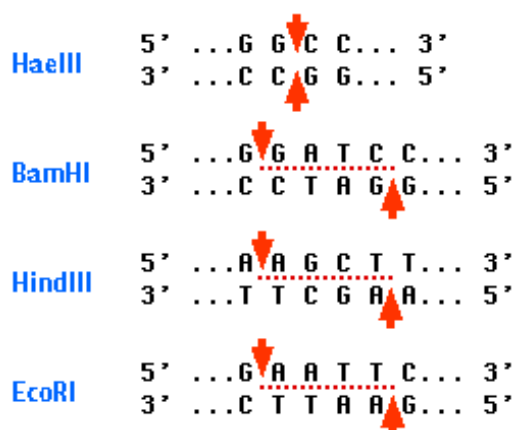
(Max 3)

**Reject: constant temperature in natural DNA replication while temperature fluctuates in PCR- not a good answer (no common basis for comparison due to different context)**

[L2]

After the target DNA has been subjected to PCR, it is cleaved using restriction enzymes and then inserted into pEOD. Fig. 3.3 shows the types of restriction enzymes that can be used to cleave pEOD at the Multiple Cloning Site (MCS) shown in Fig. 3.1

**Restriction enzyme      Specific target base sequence of DNA**



**Fig. 1.3**

**(c)(i)** State the restriction enzyme used to cleave the target DNA sequence before it is inserted into vector pEOD. [1]

1. BamHI

[L2]

(ii) Outline the steps that must subsequently be followed to produce a recombinant plasmid. [3]

| SC                                | OR                      | E                                                                                                                                                                              |
|-----------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Production of recombinant plasmid | • Restriction digestion | 1. The <u>target sequence</u> and <u>plasmid</u> are cleaved using <u>BamHI / same restriction enzyme</u> to give <u>sticky ends / staggered ends</u> .                        |
|                                   | • Annealing             | 2. The target sequence and the plasmid are then mixed together to allow <u>sticky ends/ staggered ends</u> to <u>anneal by hydrogen bonding / complementary base pairing</u> . |
|                                   | • Ligation              | 3. <u>DNA ligase</u> is added to form <u>phosphodiester bonds/seal the nicks</u> between the target sequence and the plasmid.                                                  |

[L1]

(d) Describe and explain how bacteria transformed with this recombinant plasmid are identified using selective agar media. [3]

| SC                                                      | OR                                                         | E                                                                                                                                                                                                                     |
|---------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Describe (process)<br>Selection of recombinant bacteria | • Selection for transformed bacteria                       | 1. The bacteria are first <u>plated onto nutrient agar plates</u> containing the <u>ampicillin</u> , this would <u>select for bacteria colonies transformed with plasmid/pEOD</u> (recombinant & reannealed plasmids) |
|                                                         | • Replica-plating                                          | 2. <u>Replica-plate</u> the bacteria colonies grown on the <u>ampicillin plate</u> onto <u>nutrient agar plates</u> containing <u>kanamycin</u>                                                                       |
|                                                         | • Selection of recombinant colonies                        | 3. <u>Obtain bacteria colonies able to grow on ampicillin plate but unable to grow on kanamycin plate</u> as bacteria transformed with recombinant plasmid / <u>OWTTE</u>                                             |
| Explain                                                 | • Insertional inactivation of Kan <sup>R</sup> marker gene | 4. <u>Bacteria</u> transformed with <u>recombinant plasmid</u> will be <u>unable to grow on kanamycin plate</u> due to <u>insertional inactivation of the Kan<sup>R</sup> marker gene</u> .                           |

(Max 3)

[L2]

[Total: 12]



(a) Explain the possible evolutionary processes that could have led to the many species seen in Fig 4.2. [2]

| SC                     | OR                                                                                                                                                                                                                                            | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evolutionary processes | <ul style="list-style-type: none"> <li>• <b>Adaptive radiation</b></li> <li>• due to <b>selection pressures</b> of food sources, habitat etc</li> </ul>                                                                                       | <ol style="list-style-type: none"> <li>1. Evolution of many species via <b><u>Adaptive radiation</u></b> over a relatively short period of time.</li> <li>2. due to different <b><u>selection pressures of food sources, habitat</u></b> etc</li> <li>3. <b><u>Sympatric speciation- no geographical isolation.</u></b></li> <li>4. <b><u>Prezygotic barriers- behavioural / temporal</u></b></li> <li>5. over time become <b><u>reproductively isolated</u></b></li> </ol> <p style="text-align: right;">(Max 2)</p> |
| Many Species           | <ul style="list-style-type: none"> <li>• <b>Sympatric speciation-</b> no geographical isolation.</li> <li>• prezygotic barriers- <b>behavioural / temporal /</b></li> <li>• <b>over time</b> become <b>reproductively isolated</b></li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

[L3]

(b) Explain how DNA homology is used to predict phylogeny. [2]

| SC              | OR                                                                                                                                                                                            | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1] DNA Homology | <ul style="list-style-type: none"> <li>• DNA nucleotide sequences</li> <li>• nucleic acid hybridization as e.g.</li> <li>• good comparison since most living organisms possess DNA</li> </ul> | <ol style="list-style-type: none"> <li>1. <b><u>DNA homology is the study of the degree of similarity of DNA between the two types of species</u></b> being studied.</li> <li>2. DNA homology is <b><u>useful</u></b> as <b><u>most organisms</u></b> (except Viruses) <b><u>possess DNA</u></b> and therefore is a <b><u>good level for comparison.</u></b></li> <li>3. The principle for comparison- the <b><u>higher the homology</u></b>, the <b><u>greater the number of complementary nucleotide base pairs</u></b>, the <b><u>higher the temperature needed to separate the two strands of DNA.</u></b> the higher the homology the <b><u>more related two species</u></b> are.</li> <li>4. the <b><u>phylogenetic relationship</u></b> between them can further be <b><u>determined</u></b> with the <b><u>greater degree of similarity.</u></b></li> </ol> |
| 2] Phylogeny    | <ul style="list-style-type: none"> <li>• greater degree of similarity- more related</li> <li>• less degree of similarity less related.</li> </ul>                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

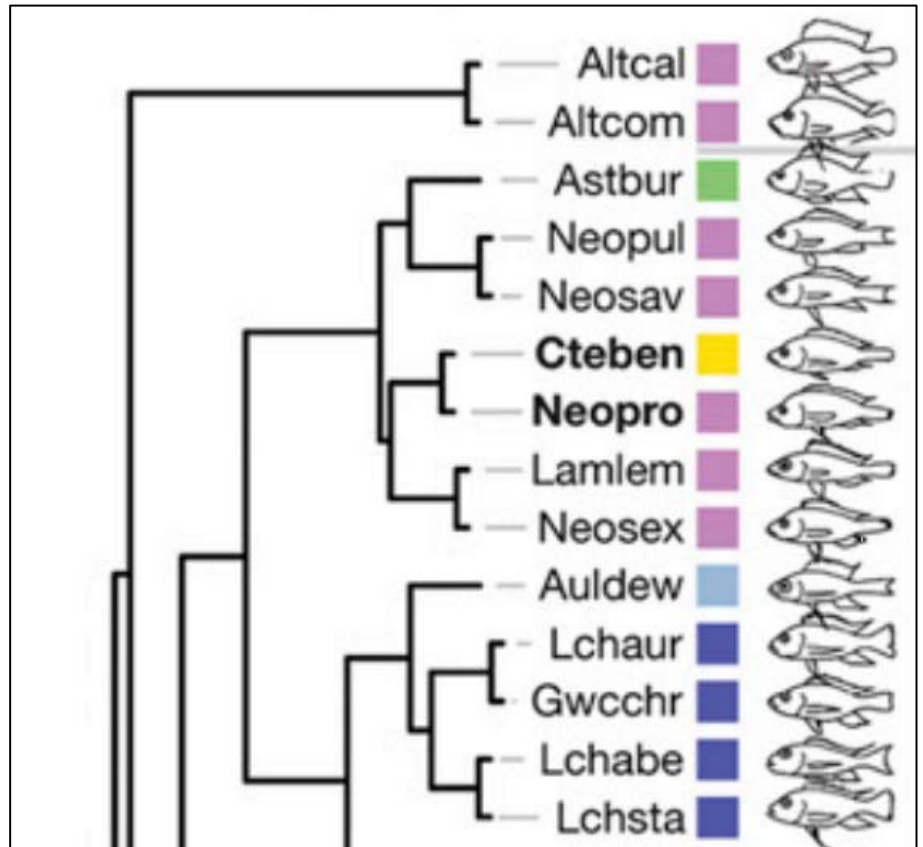
[L2]

## (A) Body Plan / Shape

**Fig 4.3**

(A) Portion of the Cluster analysis \* proposing phylogenetic relationships on the basis of 17 landmarks on body shape. Outlines in (A) are based on real photographs

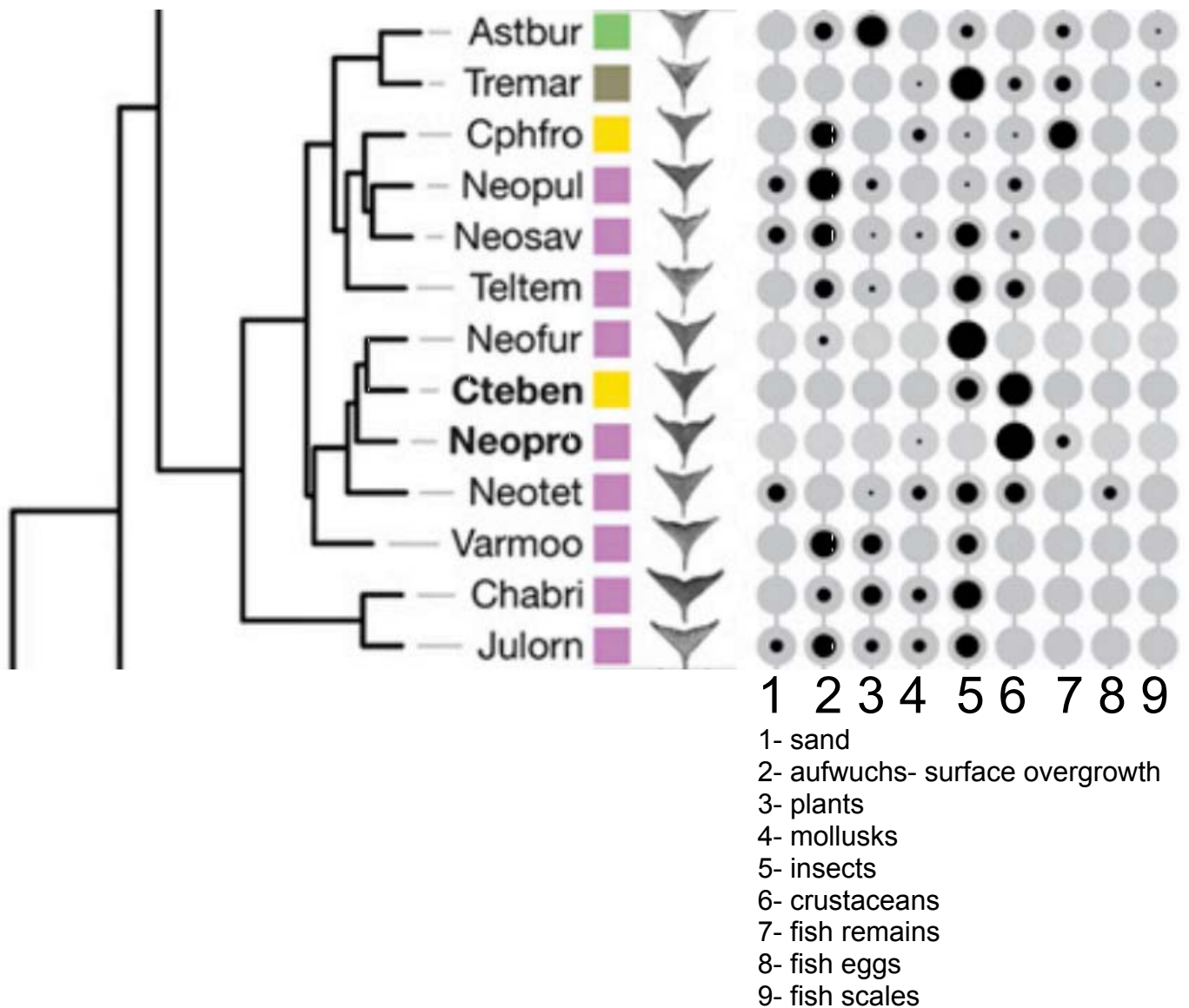
\*Cluster Analysis is a systematic comparison of parameters that allows a conclusion on relationships between specimens.



Current Biology 22, 2362–2368, December 18, 2012 ©2012 Elsevier Ltd All rights reserved <http://dx.doi.org/10.1016/j.cub.2012.10.048> adapted

(B) lower pharyngeal jaw bone

(C) gut content



Current Biology 22, 2362–2368, December 18, 2012 ©2012 Elsevier Ltd All rights reserved <http://dx.doi.org/10.1016/j.cub.2012.10.048> adapted

**Fig 4.4**

(B) Portion of the Cluster analysis proposing phylogenetic relationships on the basis of landmarks on the lower pharyngeal jaw bone. images in (B) are taken from dissected LPJs  
(C) Results from the stomach and gut content analyses (in volume %).

- (c) Explain the significance of the results found in Fig 4.2, 4.3 and 4.4 on the phylogeny of 'Neopro' and 'Cteben' species. [2]

| SC                            | OR                                                                                                                                                                                                                    | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fig 8.2<br>Phylogeny          | <ul style="list-style-type: none"> <li>• Homology of DNA</li> <li>• Neopro and Ctenbens are separate Genus Neolamprologus vs Ctenochromis</li> <li>• Neopro and Ctebens separate species [not significant]</li> </ul> | <p>Compulsory</p> <ol style="list-style-type: none"> <li>1. in Fig 4.2 comparing <u>homology of DNA</u> place Neopro and Ctebens as <u>separate Genera</u>.</li> <li>2. in Fig 4.3 comparing <u>homology of Body plans</u> place Neopro and Ctebens as more closely related arising from the <u>same speciation event / i.e. same genus</u></li> <li>3. in Fig 4.4 comparing <u>homology of jaw bones</u> place Neopro as a <u>common ancestor</u> to Ctebens / i.e. <u>same genus</u>.</li> </ol> <p>Compulsory</p> <ol style="list-style-type: none"> <li>4. Results in Fig 4.3 and Fig 4.4 are in contrast to Fig 4.2. suggesting that Neopro and Ctebens are more <u>closely related</u>.</li> <li>5. Possible <u>convergent evolution</u></li> <li>6. possibly due to the <u>same selection pressures of food sources- crustaceans</u>.</li> </ol> <p>(Max 2)</p> |
| Fig 8.3<br>Body Plan          | <ul style="list-style-type: none"> <li>• Homology in Body plans suggest they are more closely related</li> <li>• suggests that they arise from a common speciation point</li> </ul>                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Fig 8.4<br>Jaw Bone / Stomach | <ul style="list-style-type: none"> <li>• Homology of Jaw bones – Neopro was common ancestor to Ctebens</li> <li>• same genus?</li> </ul>                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Significance                  | <ul style="list-style-type: none"> <li>• Possible convergent evolution</li> <li>• due to same selection pressure- food source- crustaceans</li> <li>• not interbreeding</li> </ul>                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

[L3]

[Total: 6]

## Section B

Answer **ONE** questions. Answer each part on a **separate** piece of paper.

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

**5(a)** Describe how carbohydrates are formed in the Calvin cycle. (ACJC(H1)/2/6) [7]

| SC                                       | OR                                                  | E                                                                                                                                                                                                                                                               |
|------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calvin cycle<br>(Light independent rxns) | • Carbon fixation                                   | 1. <u>CO<sub>2</sub> combines with 5C sugar ribulose bisphosphate (RuBP);</u><br>2. <u>Carbon fixation</u> catalysed by <u>RuBP carboxylase oxygenase/Rubisco;</u>                                                                                              |
|                                          | 1. Carbon reduction                                 | 3. <u>6C intermediate splits to form two glycerate-3-phosphate (GP);</u><br>4. <u>use of ATP (in carbon reduction);</u><br>5. <u>reduction / use of NADPH (in carbon reduction);</u><br>6. to form <u>1 glyceraldehyde-3-phosphate (GALP/triose phosphate);</u> |
|                                          | 2. RuBP regeneration + biosynthesis of carbohydrate | 7. <u>GALP/G3P used to regenerate RuBP,</u><br>8. utilising <u>ATP</u> molecules ( <u>in RuBP regeneration</u> );<br>9. <u>GALP/G3P/triose phosphate</u> used for <u>synthesis of carbohydrate;</u>                                                             |

[L1]

**(b)** Outline the role of NAD and NADP in cells. (N2005/2/7a) [7]

| SC                                                                       | OR                                                 | E                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Role<br>(Function)<br><br>NAD → respiration<br><br>NADP → Photosynthesis | NAD/NADP – electron carrier                        | 1. NADP/ NAD as <u>electron/hydrogen carrier</u> or <u>coenzymes/</u> carry hydrogen atoms from one compound to another;                                                                                                                                                                                   |
|                                                                          | NAD in respiration<br><br>Aerobic<br><br>Anaerobic | Aerobic respiration:<br>2. Glycolysis: <u>Accept H<sup>+</sup> from triose phosphate/ glyceraldehyde phosphate, which is oxidized to 1,3 diphosphoglycerate or pyruvate;</u><br>3. <u>Forms 2 NADH for each glucose molecule;</u><br>4. Link reaction: <u>Accept H<sup>+</sup> from pyruvate,</u> which is |

|  |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |                        | <p><u>converted to acetyl CoA</u>;</p> <p>5. Forms <u>2 NADH</u> for each glucose molecule (<u>link reaction</u>);</p> <p>6. <u>Krebs cycle</u>: <u>Accept H<sup>+</sup> from intermediates</u> in the regeneration of oxaloacetate from citrate;</p> <p>7. Forms <u>6 NADH</u> for each glucose molecule (<u>Krebs cycle</u>);</p> <p>8. <u>Link rxn</u> and <u>Krebs cycle occurs in the mitochondrial matrix</u>;</p> <p>9. <u>Carry hydrogen/ electrons</u> to <u>ETC</u>;</p> <p>10. <u>Oxidative phosphorylation/ transfer of electrons down the ETC</u> to release ATP;</p> <p>11. <u>1 NADH yields 3/2.5 ATP</u> molecules;</p> <p>12. <u>NADH reoxidised</u> in <u>oxidative phosphorylation</u>;</p> <p>Anaerobic respiration:</p> <p>13. <u>NAD regenerated</u> when <u>electrons from NADH transferred to pyruvate/ ethanal</u>;</p> <p>14. <u>NAD reused in glycolysis</u> to <u>generate 2 ATP</u>;</p> |
|  | NADP in photosynthesis | <p>15. <u>Final electron acceptor</u> in <u>non-cyclic photophosphorylation</u>;</p> <p>16. <u>Drives photolysis of water</u>;</p> <p>17. <u>2H<sup>+</sup> + 2e<sup>-</sup> + NADP → NADPH</u> or reduced NADP/ hydrogen ions from splitting of water</p> <p>18. <u>combine with NADP</u> to form <u>NADPH</u>;</p> <p>19. <u>NADPH passes to Calvin cycle</u>/ light-independent reactions in the stroma;</p> <p>20. <u>NADPH reduces glycerate biphosphate/1,3 diphosphoglycerate</u> <u>to triose phosphate</u>;</p> <p>21. <u>NADPH</u> gets <u>reoxidized back</u> to <u>NADP</u>;</p>                                                                                                                                                                                                                                                                                                                          |

[L2]

- (c) Compare and contrast the process of electron transport through the electron transport chain in the mitochondrion and chloroplast. (Novel) [6]

| SC                                                                                                                       | OR                                                                                                                                                                                                                                                                         | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Compare → similarity &amp; difference</p> <p>Context → electron transport in ETC of mitochondrion and chloroplast</p> | <p>Similarities:</p> <ul style="list-style-type: none"> <li>• e carriers organized in decreasing energy levels.</li> <li>• Energy released pump H<sup>+</sup></li> <li>• Proton gradient</li> <li>• ATP synthesis by ATP synthase</li> <li>• Location: Membrane</li> </ul> | <p>1. <u>Series of electron carriers / cytochrome</u> are of <u>decreasing energy levels</u>;</p> <p>2. <u>Energy released</u> is used to <u>actively pump H<sup>+</sup></u> across membrane;</p> <p>3. to create a <u>H<sup>+</sup> gradient / proton motive force</u>;</p> <p>4. Which is used to <u>drive ATP synthesis</u> as <u>H<sup>+</sup> return via Stalk particle containing ATP synthase</u> down the H<sup>+</sup> gradient / <u>chemiosmosis</u>;</p> <p>5. <u>Both ETCs</u> are <u>found</u> on <u>membranes of organelles</u>.</p> |

|  |                                                             |                                                                              |                                                                                                   |
|--|-------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|  |                                                             | (Max 3)                                                                      |                                                                                                   |
|  | Differences (basis of comparison) – [Max 3]:                | <ul style="list-style-type: none"> <li>ETC in Mitochondrion</li> </ul>       | <ul style="list-style-type: none"> <li>ETC in Chloroplast</li> </ul>                              |
|  | 6. Transfer of electrons to ETC                             | From <u>NADH</u> and <u>FADH<sub>2</sub></u> ; (reject coenzyme)             | From <u>reaction centre / chlorophyll a</u> / electron acceptor of photosystems / primary pigment |
|  | 7. Final electron acceptor                                  | <u>Oxygen</u>                                                                | <u>NADP<sup>+</sup></u>                                                                           |
|  | 8. Products formed                                          | <u>H<sub>2</sub>O</u> , <u>ATP</u> , <u>NADH</u> and <u>FADH<sub>2</sub></u> | <u>NADPH</u> , <u>ATP</u> and <u>O<sub>2</sub></u>                                                |
|  | 9. Location                                                 | <u>Inner mitochondrion membrane</u>                                          | <u>Thylakoid membrane</u>                                                                         |
|  | 10.Direction of proton pumping                              | <u>From matrix into intermembranal space</u>                                 | <u>From stroma into thylakoid lumen / thylakoid space</u>                                         |
|  | 11.Direction of proton movement down concentration gradient | <u>From intermembrane space into matrix</u>                                  | <u>From thylakoid space into stroma</u>                                                           |
|  | 12.Nature of ETC                                            | Electron transport is <u>non-cyclic</u>                                      | Electron transport can be either <u>cyclic</u> or <u>non-cyclic</u>                               |
|  | 13.Source of energy                                         | <u>Breakdown of glucose</u>                                                  | <u>Light energy</u>                                                                               |
|  | 14.Nature of metabolic process                              | Involved in <u>catabolic process</u>                                         | Involved in <u>anabolic process</u>                                                               |
|  | 15.No. of ETC                                               | <u>One</u> in cellular respiration                                           | <u>Two</u> in non-cyclic photophosphorylation                                                     |

[L3]

**6(a)** Describe the main characteristics as well as the different developmental potential of stem cells. Cite examples wherever appropriate. [7]

| SC                                                       | OR                                                                      | E                                                                                     |
|----------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| CW: Describe<br><br>QKW: main characteristics, stem cell | <ul style="list-style-type: none"> <li>Unspecialised cell</li> </ul>    | 1. A stem cell is an <u>unspecialised cell</u>                                        |
|                                                          | <ul style="list-style-type: none"> <li>Able to differentiate</li> </ul> | 2. which has the ability to <u>differentiate</u> into a <u>specialised cell</u> type. |

|  |                                                                               |                                                                                                                                                          |
|--|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>• Conditions</li> </ul>                | 3. In the presence of <u>appropriate growth factors</u> and <u>nutrients</u> .                                                                           |
|  | <ul style="list-style-type: none"> <li>• Self-renew</li> </ul>                | 4. A stem cell is also able to <u>self-renew</u> i.e. produce more undifferentiated cells after cell division                                            |
|  | <ul style="list-style-type: none"> <li>• Proliferate continually</li> </ul>   | 5. as well as <u>proliferate continually</u> .                                                                                                           |
|  | <ul style="list-style-type: none"> <li>• Totipotent</li> </ul>                | 6. <u>Totipotent stem cells</u> are stem cells that are able to <u>differentiate into any cell type of the body including extraembryonic tissues</u> .   |
|  | <ul style="list-style-type: none"> <li>• Zygotic stem cells</li> </ul>        | 7. An example is <u>zygote</u> , as well as <u>cells within the 2, 4 &amp; 8-cell embryos</u> .                                                          |
|  | <ul style="list-style-type: none"> <li>• Pluripotent</li> </ul>               | 8. <u>Pluripotent stem cells</u> are stem cells that are able to <u>differentiate into any cell type of the body except the extraembryonic tissues</u> . |
|  | <ul style="list-style-type: none"> <li>• Embryonic stem cells</li> </ul>      | 9. An example is the <u>embryonic stem cell</u> that is found in the <u>inner cell mass</u> of the <u>blastocyst</u> .                                   |
|  | <ul style="list-style-type: none"> <li>• Multipotent stem cells</li> </ul>    | 10. <u>Multipotent stem cells</u> are stem cells that are able to <u>differentiate into limited cell types of the body</u> .                             |
|  | <ul style="list-style-type: none"> <li>• Haematopoietic stem cells</li> </ul> | 11. An example is <u>haematopoietic stem cells</u> that is found in the <u>bone marrow</u> .                                                             |

(Max 7)

[L1]

- (b) Describe how bacteria can be used in genetic engineering to improve the nutritional content of crops. (Novel) [7]

| SC                                                                                                                    | OR                                                                              | E                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>CW: Describe with an example</p> <p>QKW: genetic engineering methodology, improve nutritional content of crops</p> | <ul style="list-style-type: none"> <li>• Two foreign genes obtained.</li> </ul> | <p>1. Two foreign genes, <u>phytoene synthase (Psy)</u> from <u>daffodil</u> and <u>Crt1</u> from <u>Erwinia uredovora</u> bacterium are <u>cloned into Ti plasmid</u>.</p> |

|  |                                                                                                   |                                                                                                                                                                                                |
|--|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>• Role of these genes.</li> </ul>                          | 2. These genes are <u>enzymes</u> that are <u>involved</u> in the <u>synthesis of <math>\beta</math>-carotene</u> , which will in turn be <u>converted</u> to <u>Vitamin A</u> .               |
|  | <ul style="list-style-type: none"> <li>• Promoter</li> </ul>                                      | 3. The <u>genes</u> are <u>placed after an endosperm-specific promoter</u>                                                                                                                     |
|  | <ul style="list-style-type: none"> <li>• No <math>\beta</math>-carotene in rice grains</li> </ul> | 4. As the <u>endosperm</u> which will go on to <u>develop into rice grains do not usually produce <math>\beta</math>-carotene</u> , there will be <u>no Vitamin A</u> present.                 |
|  | <ul style="list-style-type: none"> <li>• Transform into bacteria</li> </ul>                       | 5. The <u>Ti plasmid harbouring the foreign genes</u> are <u>transformed into <i>Agrobacterium tumefaciens</i></u> .                                                                           |
|  | <ul style="list-style-type: none"> <li>• Infection &amp; transfer of genes</li> </ul>             | 6. <u>Rice embryo</u> is <u>infected</u> with the <u><i>Agrobacterium tumefaciens</i></u> bacteria and the <u>plasmids</u> carrying the gene will be <u>transferred into the plant cells</u> . |
|  | <ul style="list-style-type: none"> <li>• Growth into mature plant</li> </ul>                      | 7. The <u>rice embryo</u> will <u>grow up into a mature plant</u> that <u>produces rice grains</u> with <u>high vitamin A</u> content                                                          |

[L2]

(c) Discuss the negative consequences of genetically-modifying organisms. (Novel)

[6]

| SC                                                                                                                                         | OR                                                                          | E                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>CW: Discuss</p> <p>QKW: negative consequences, genetically-modified crop organisms (can be plants and animals)</p> <p>CON (animals)</p> | <ul style="list-style-type: none"> <li>• Competition</li> </ul>             | 1. The genetically-modified organism may <u>escape into the wild</u> and as a result of certain selective advantageous traits may <u>outcompete</u> naturally-occurring organisms <u>for resources</u> . |
| CON (animals)                                                                                                                              | <ul style="list-style-type: none"> <li>• Reduction of population</li> </ul> | 2. This competition may hence <u>reduce</u> or <u>wipe out naturally-occurring organisms</u> .                                                                                                           |
| CON (animals)                                                                                                                              | <ul style="list-style-type: none"> <li>• Debate</li> </ul>                  | 3. <u>Animal rights groups</u> and certain <u>religious groups</u> are strongly <u>opposed</u> to genetically modifying organisms and may influence the general public to avoid GMOs.                    |

|             |                                                                                                |                                                                                                                                                                                                                                                                                                                          |
|-------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CON (plant) | <ul style="list-style-type: none"> <li>• More invasive plants</li> <li>• Superweeds</li> </ul> | 4. The introduced gene(s) may be <u>transferred by pollen</u> to <u>wild relatives</u> whose hybrid offspring will become more <u>invasive /</u> become <u>'superweeds'</u> .                                                                                                                                            |
| CON (plant) | <ul style="list-style-type: none"> <li>• Cost involved in removing superweeds</li> </ul>       | 5. This may lead to <u>additional cost</u> for the <u>removal</u> of such <u>'superweeds'</u> .                                                                                                                                                                                                                          |
| CON (plant) | <ul style="list-style-type: none"> <li>• Plant diversity compromised</li> </ul>                | 6. 'Superweeds' may also <u>reduce plant biodiversity</u> by <u>out-competing</u> natural plants                                                                                                                                                                                                                         |
| CON (plant) | <ul style="list-style-type: none"> <li>• Organic farms may be compromised</li> </ul>           | 7. The introduced gene(s) may be <u>transferred by pollen</u> to <u>unmodified plants</u> growing on a farm with <u>'organic'</u> certification.                                                                                                                                                                         |
| CON (plant) | <ul style="list-style-type: none"> <li>• Toxic components</li> </ul>                           | <p>8. The modified plants will be a direct hazard to humans, domestic animals or other beneficial animals by being <u>toxic</u>.</p> <p>OR</p> <p>For instance, the herbicides that can now be used on the crop will itself <u>leave toxic residues</u> in the crop, which will be toxic to humans who consume them.</p> |
| CON (plant) | <ul style="list-style-type: none"> <li>• Damage environment</li> </ul>                         | 9. The toxic residues may also affect <u>and damage the environment</u> by causing pollution to the surroundings.                                                                                                                                                                                                        |
| CON (plant) | <ul style="list-style-type: none"> <li>• Allergies</li> </ul>                                  | 10. The modified plants will be a direct hazard to humans, domestic animals or other beneficial animals by being <u>producing allergies</u> upon consumption.                                                                                                                                                            |
| CON (plant) | <ul style="list-style-type: none"> <li>• Farmers pay royalties</li> </ul>                      | <p>11. Farmers may have to pay royalties to sow the crops.</p> <p>OR</p> <p>Some companies have extended their <u>patents on genetically engineered seeds</u> to prevent farmers from re-sowing the seed from these genetically engineered crops.</p>                                                                    |

|             |                                                                        |                                                                                                                                                  |
|-------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| CON (plant) | <ul style="list-style-type: none"> <li>Higher consumer cost</li> </ul> | 12. The cost that farmers have to pay for the royalties may be passed on the <u>consumers</u> who may have to <u>pay higher prices</u> for them. |
|-------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|

[L3]

END OF PAPER